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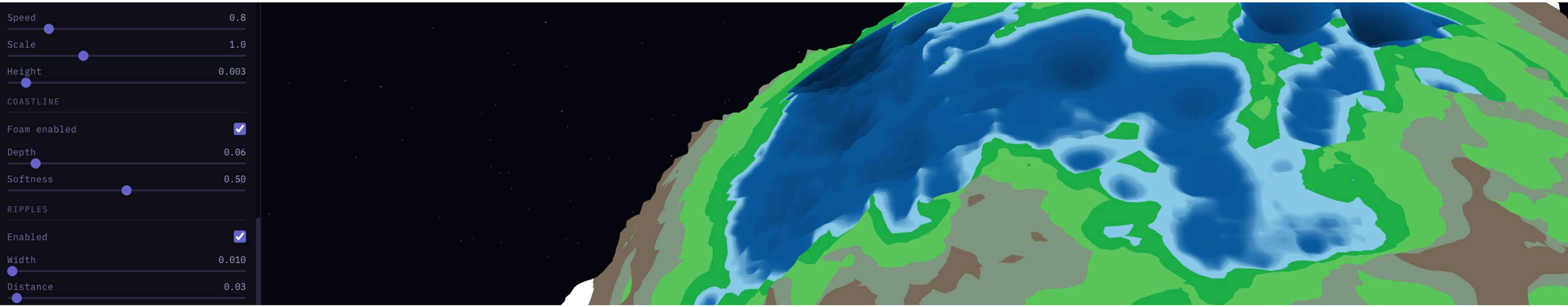

Procedural World Building

By Jose Sanchez
Design Tech - 2026

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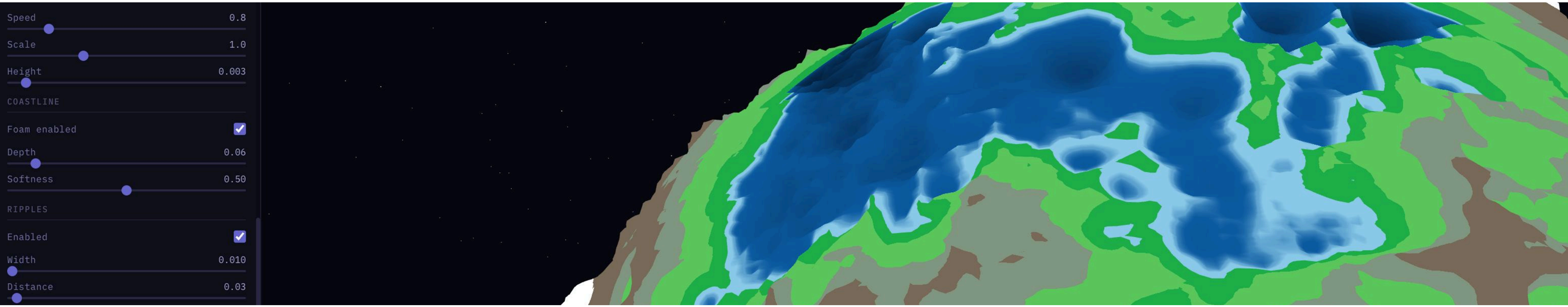

Class 01

DESIGN 6197/4197 - Fall 2026

Jose Sanchez
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About me
Jose Sanchez

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This course:
Procedural World Building

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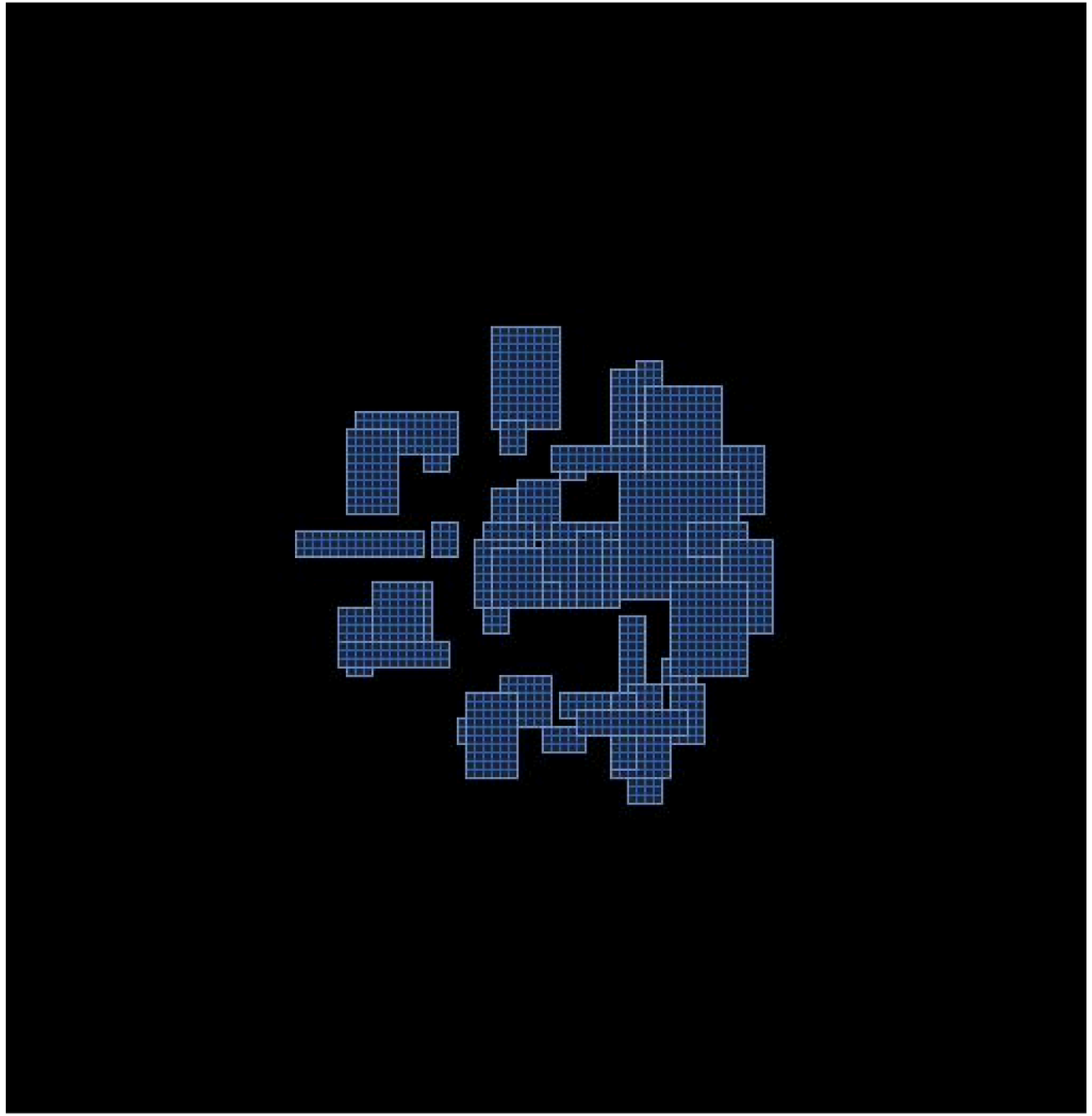

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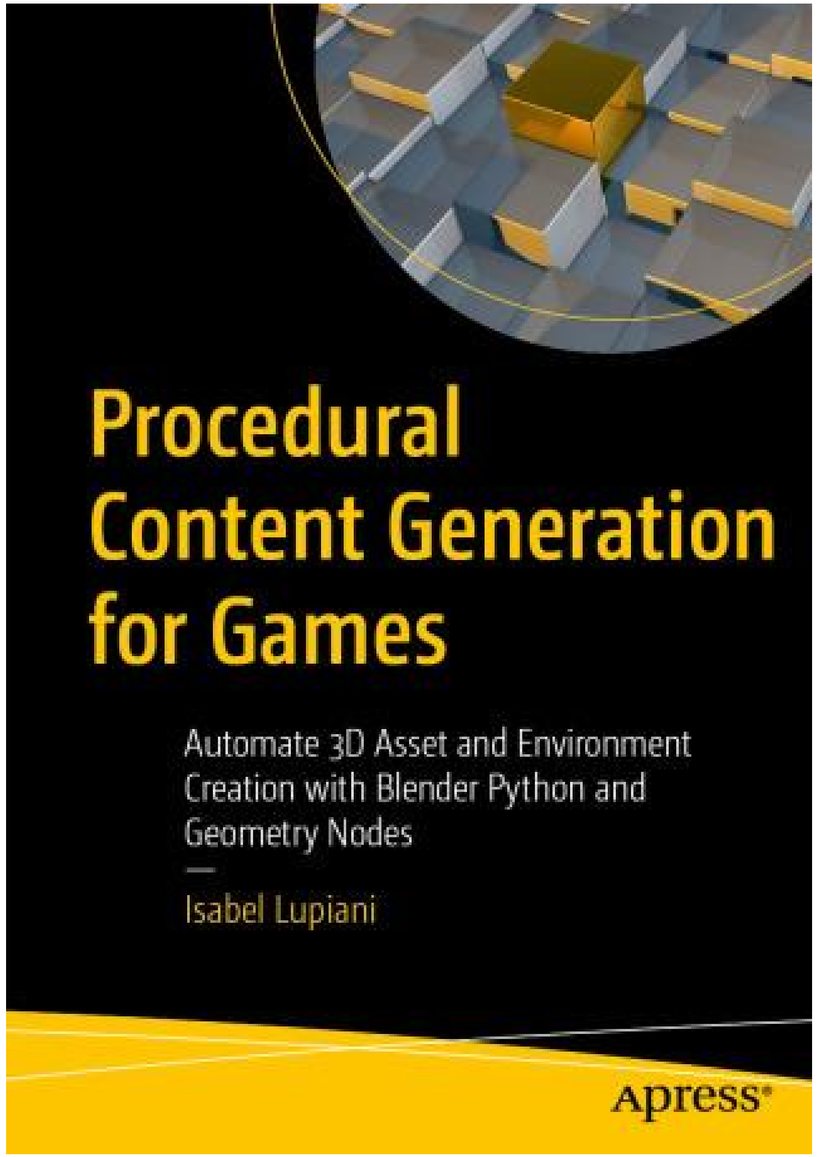

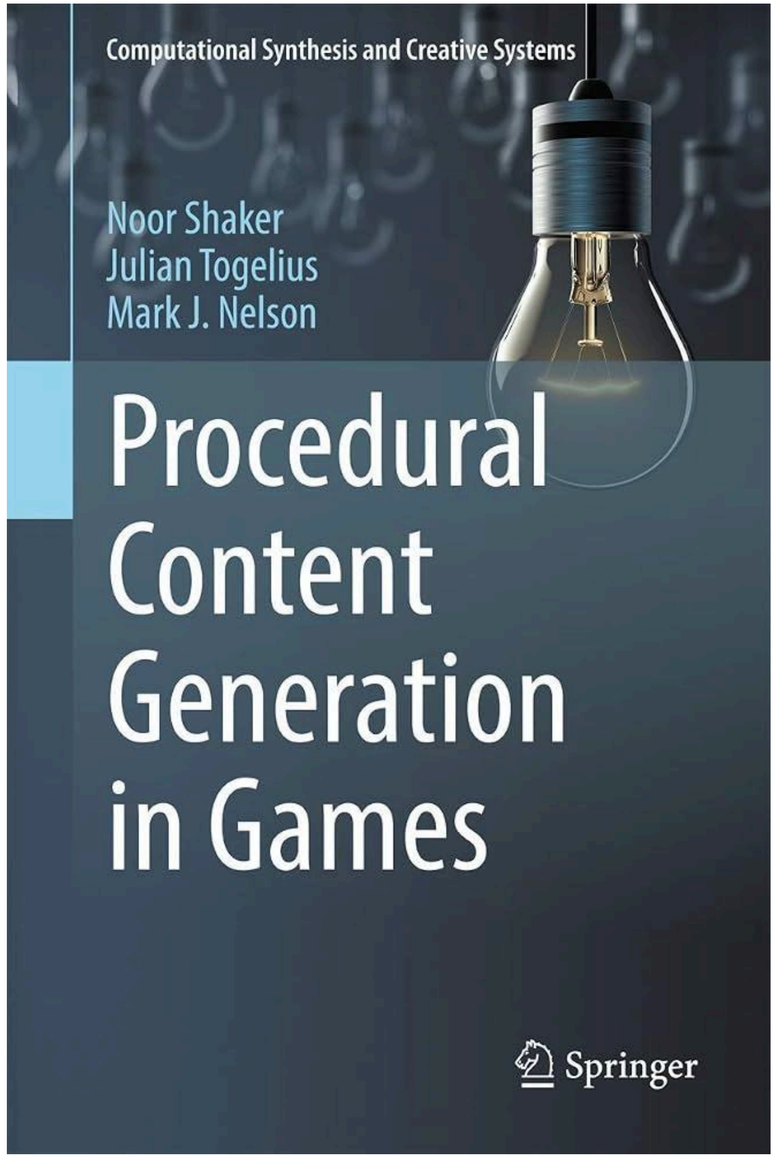

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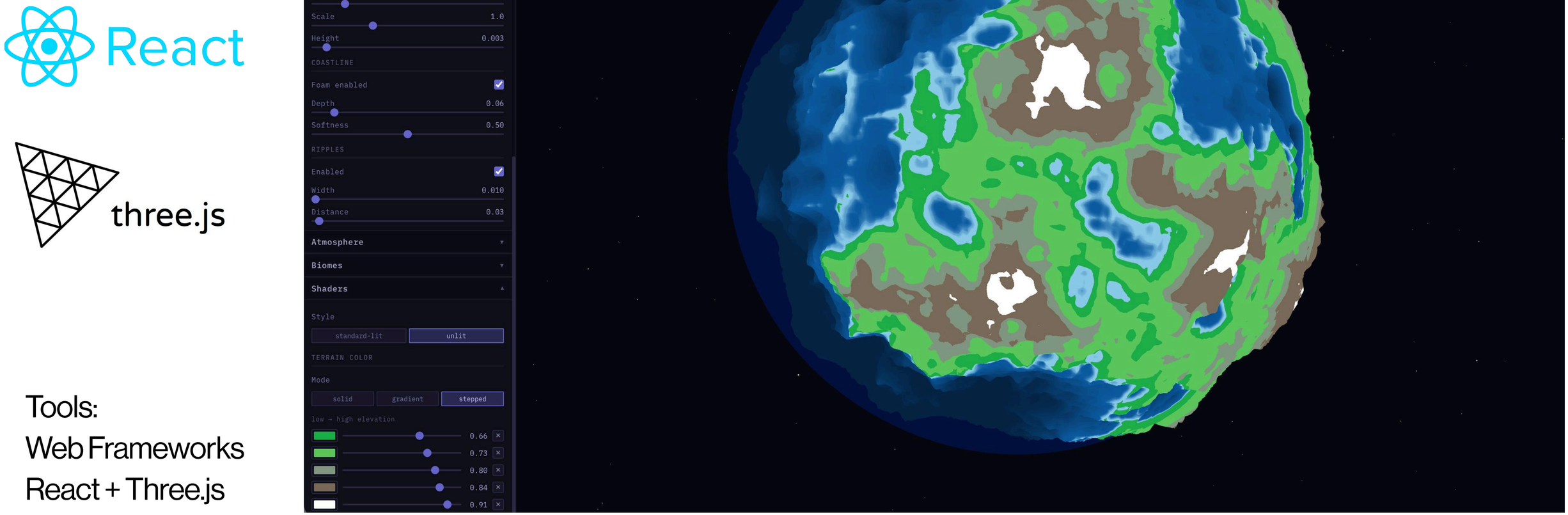

Tools:
Web Frameworks
React + Three.js

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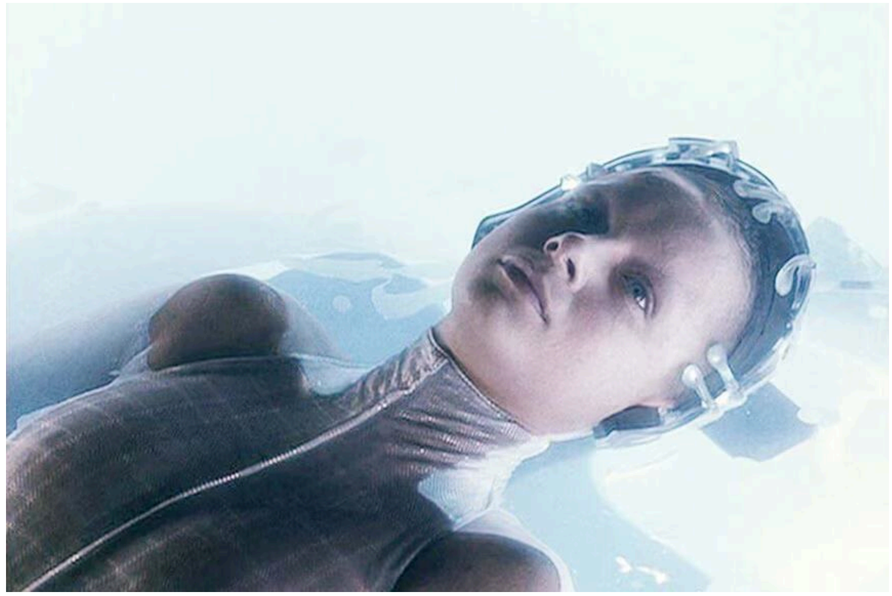

World Building
Alex McDowell

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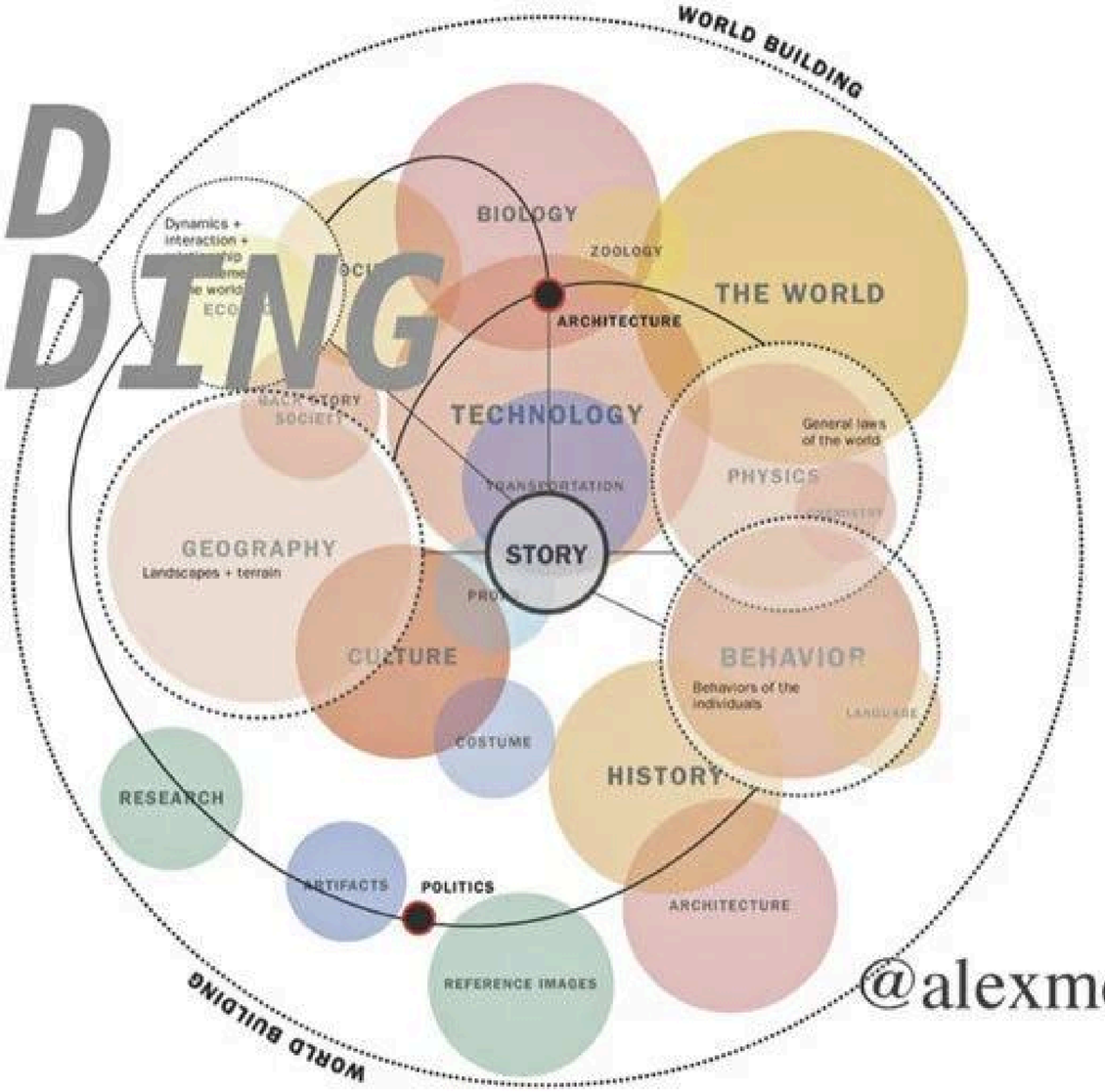
WORLD BUILDING



@alexmcdowell

World Building
Alex McDowell

Courtesy Alex McDowell, USC 5D Institute

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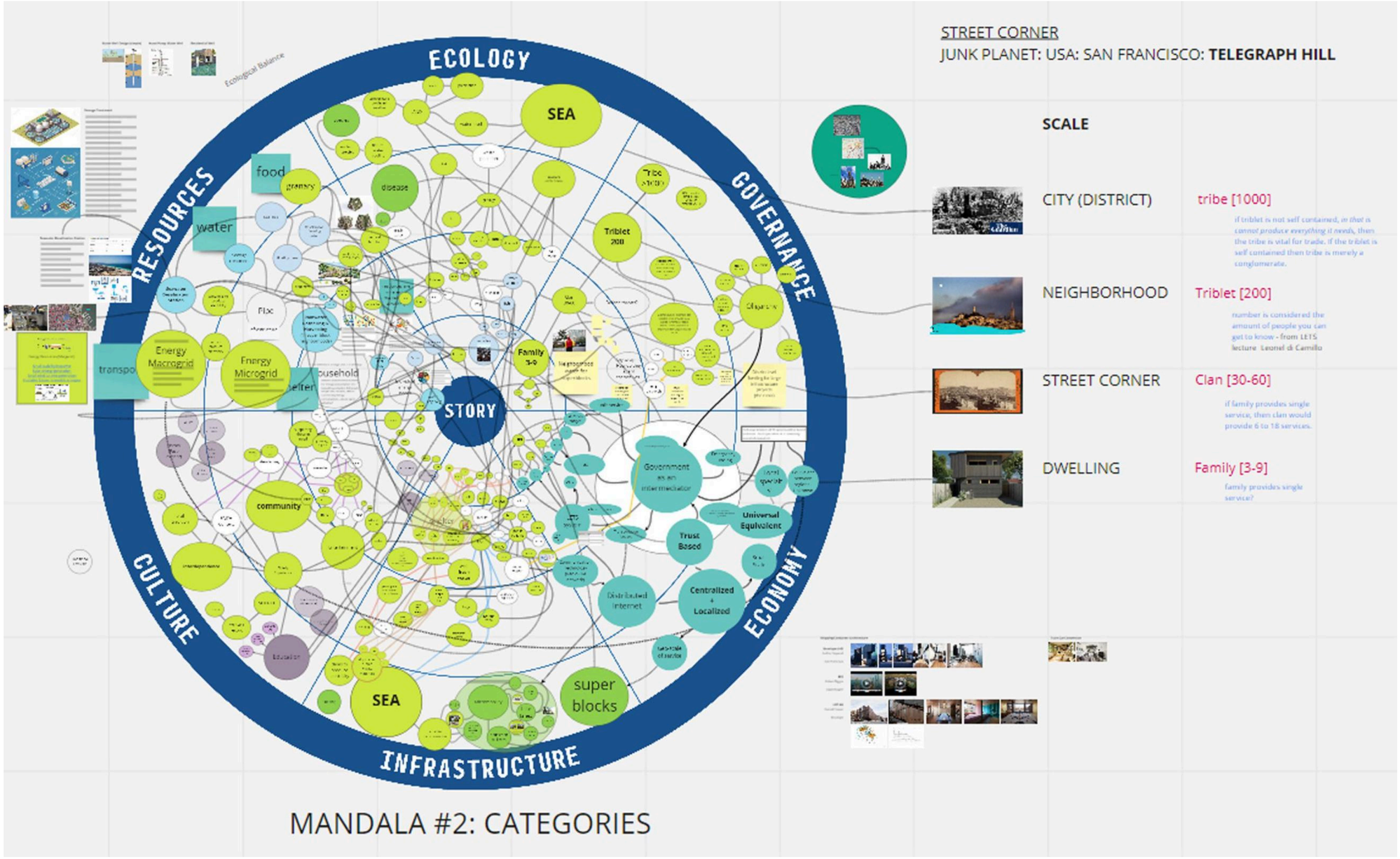
World Building
Alex McDowell



MANDALA #2: CATEGORIES

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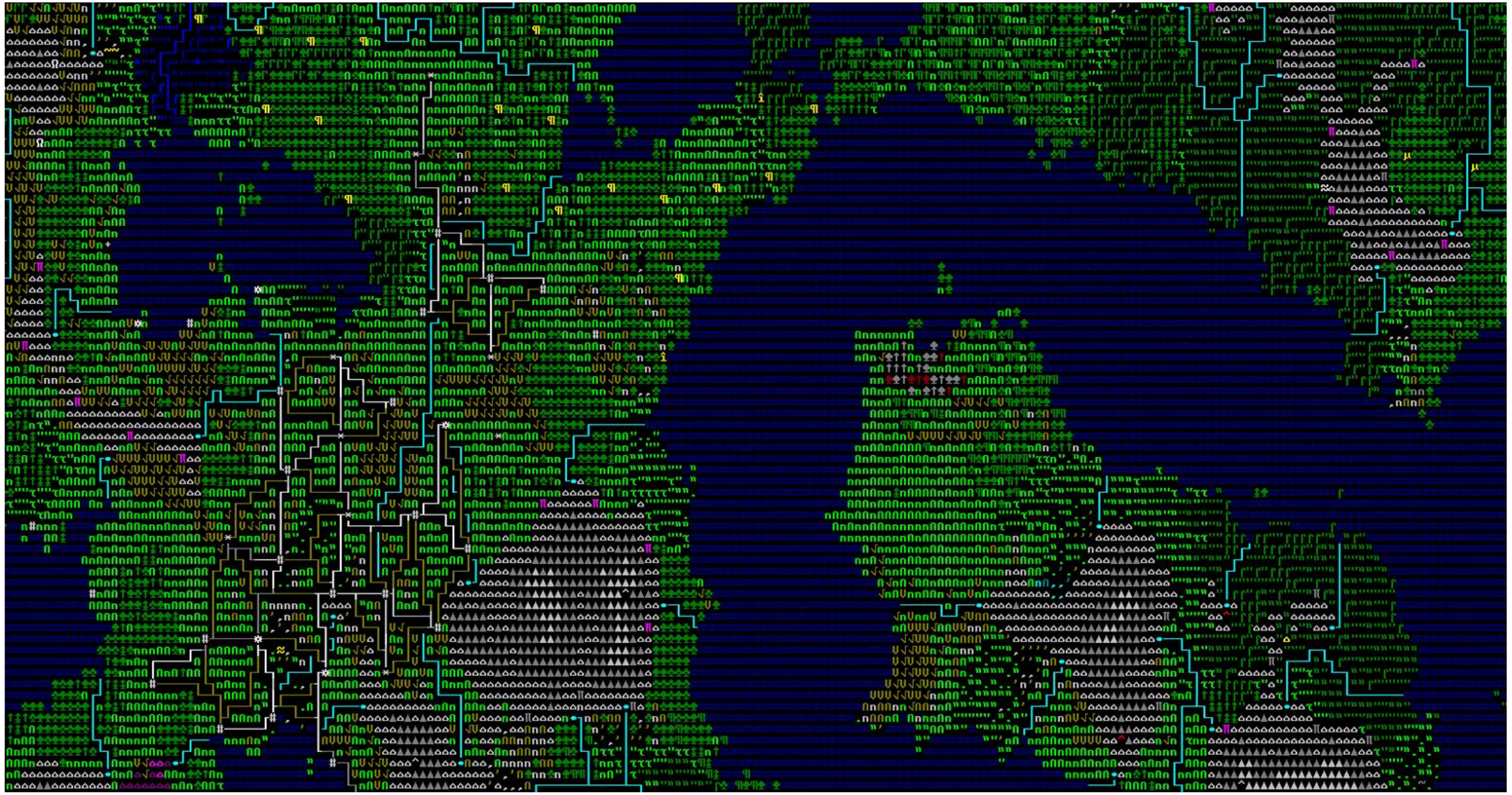
World Building



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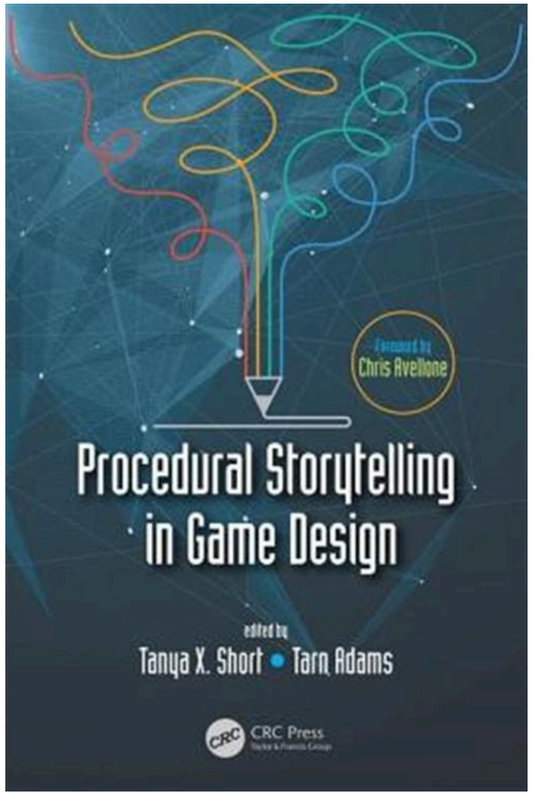

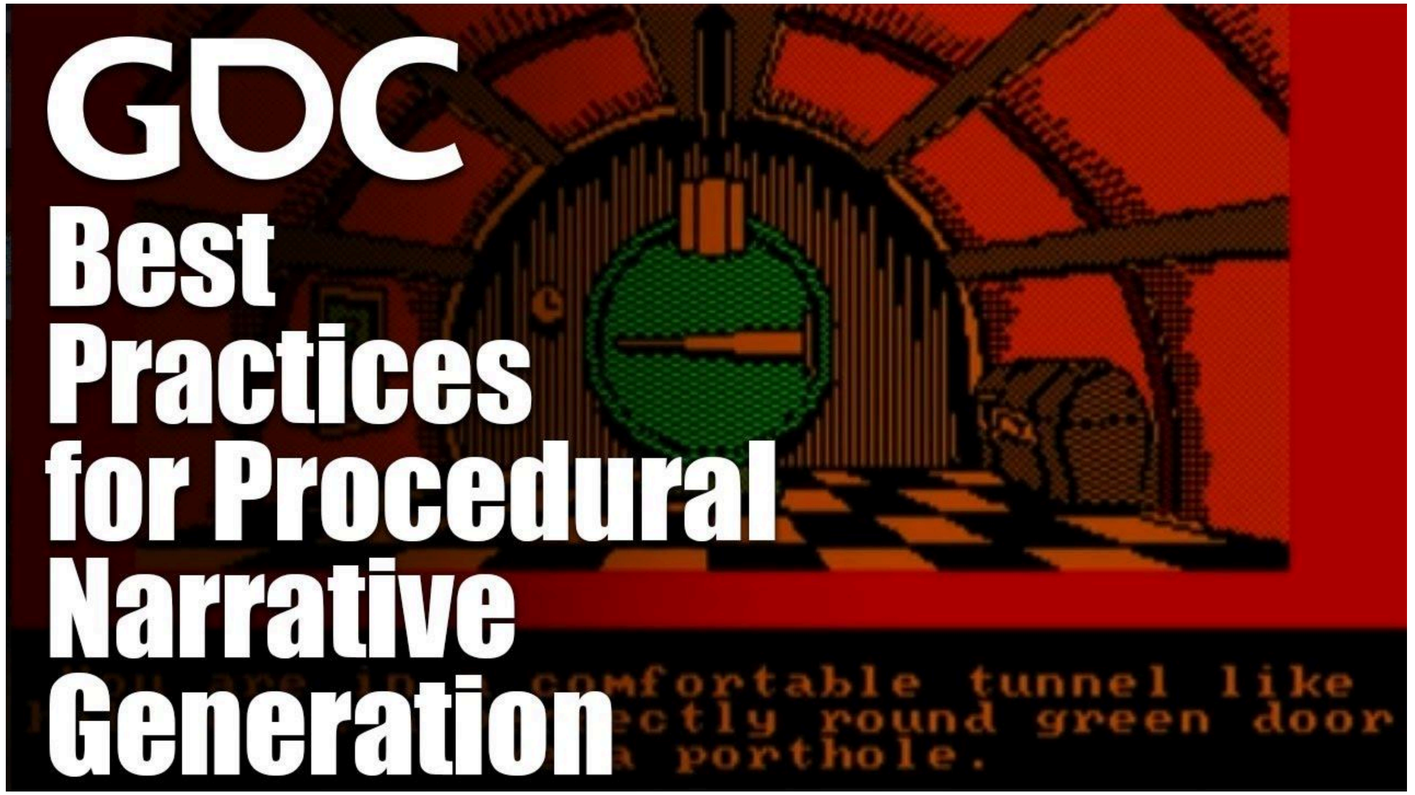

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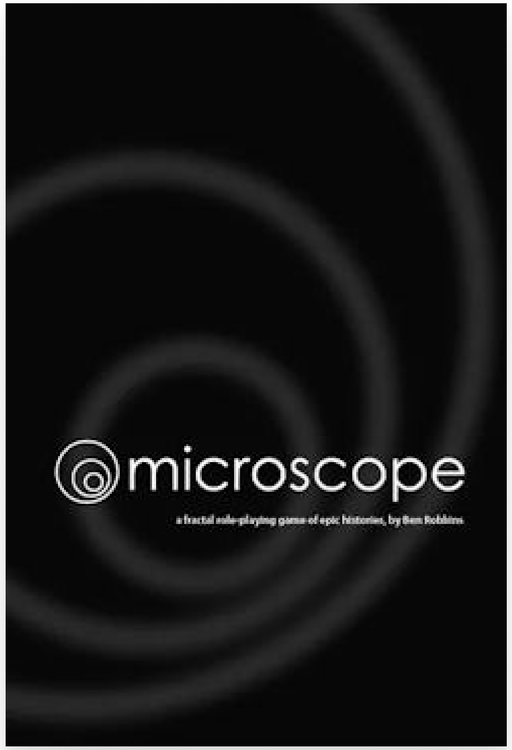

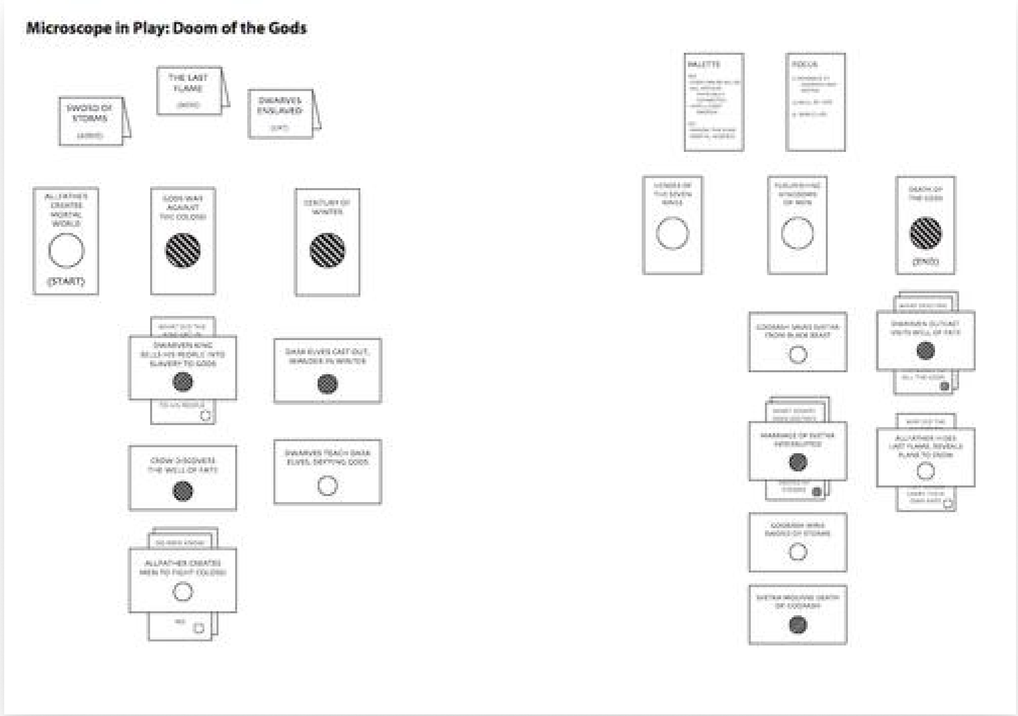

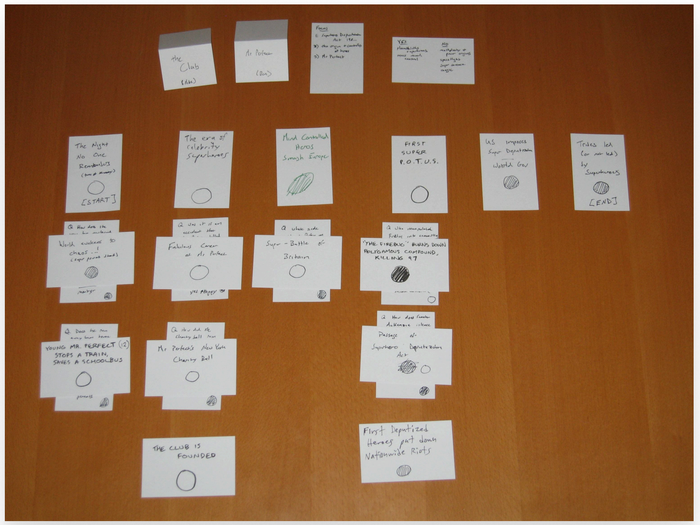

World Building
Microscope RPG

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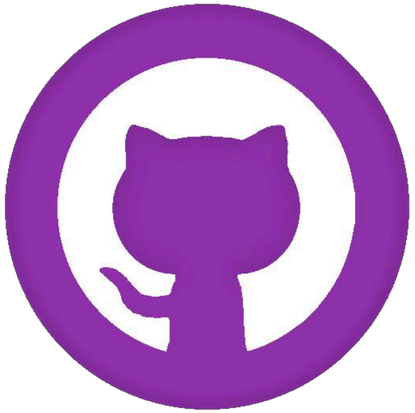
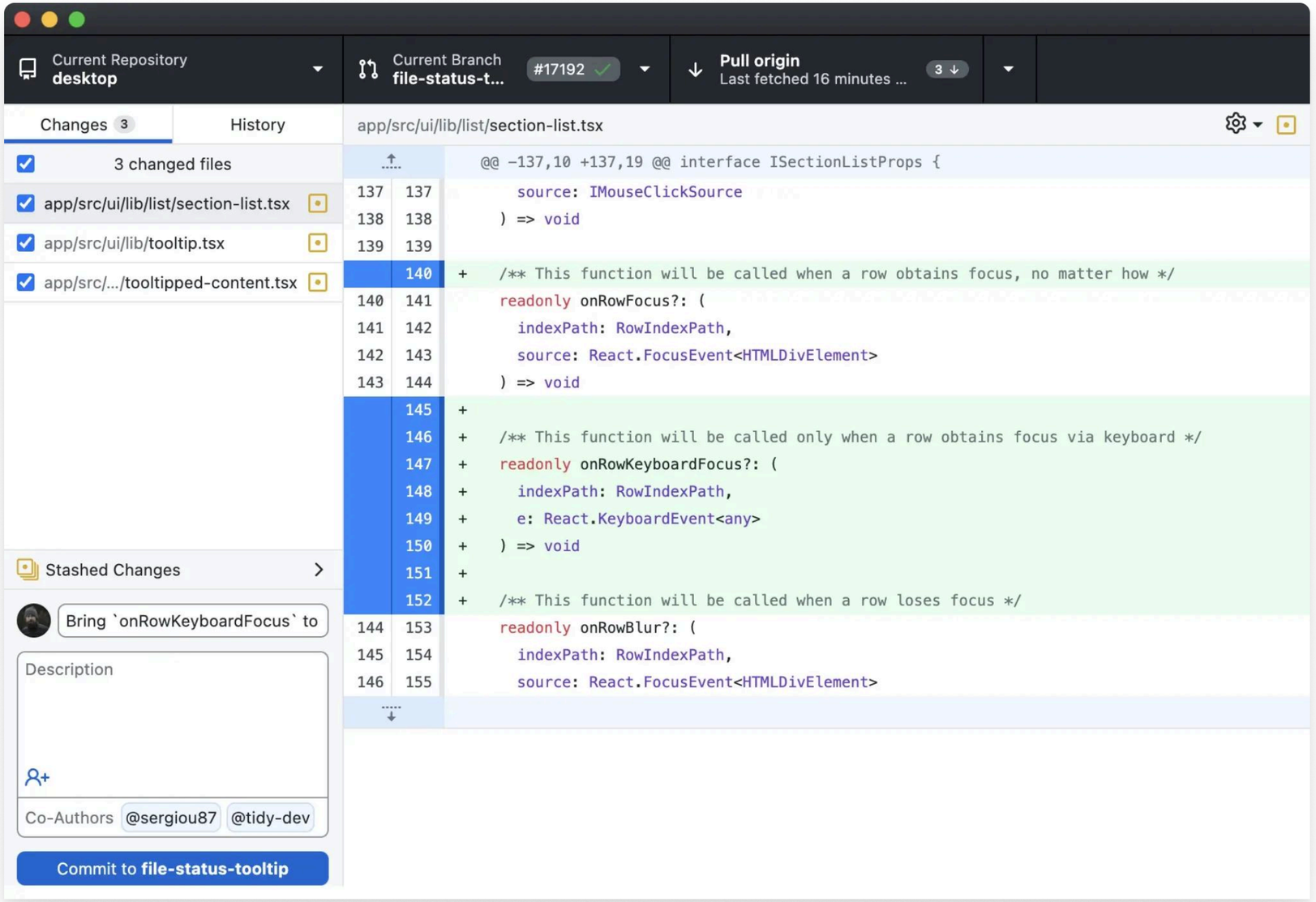


Tools
Version Control
Git - Github



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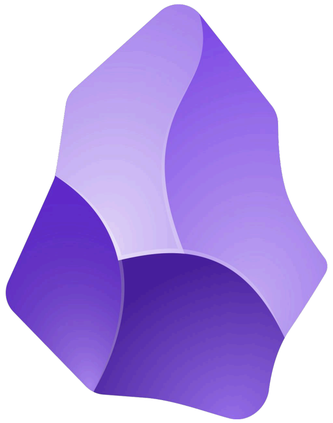
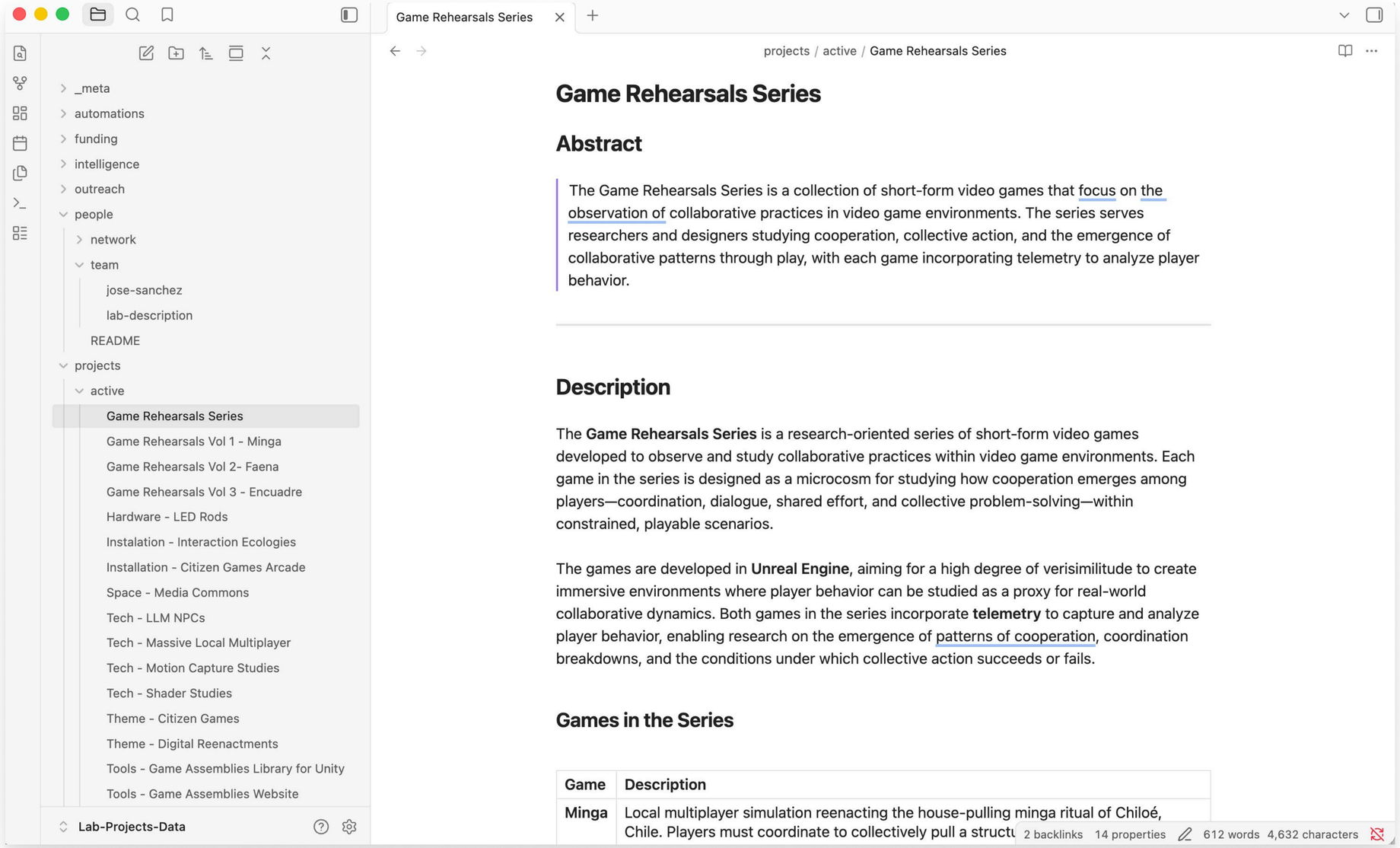


Tools
Markdown Notes



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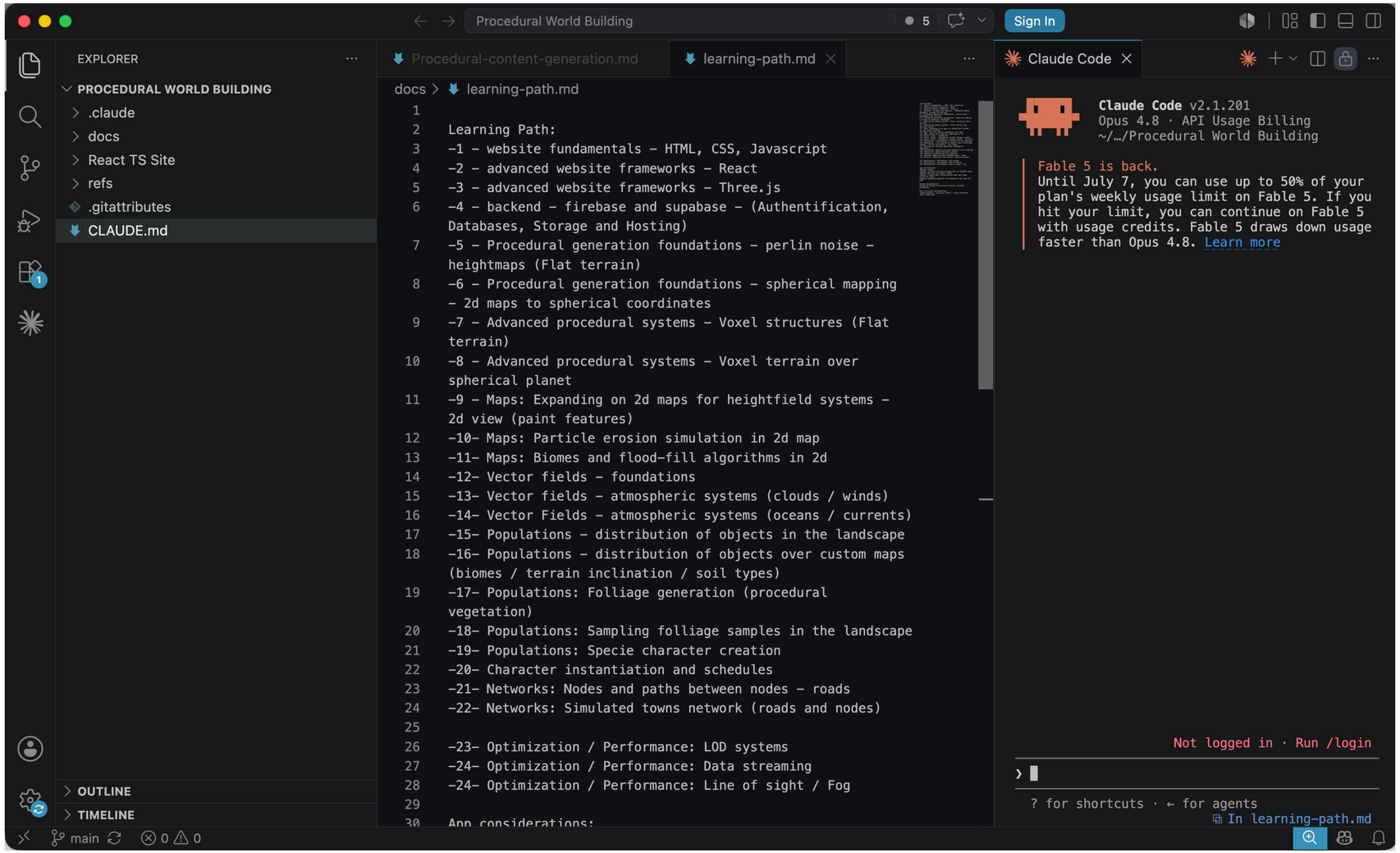

Tools
Claude Code
Visual Studio

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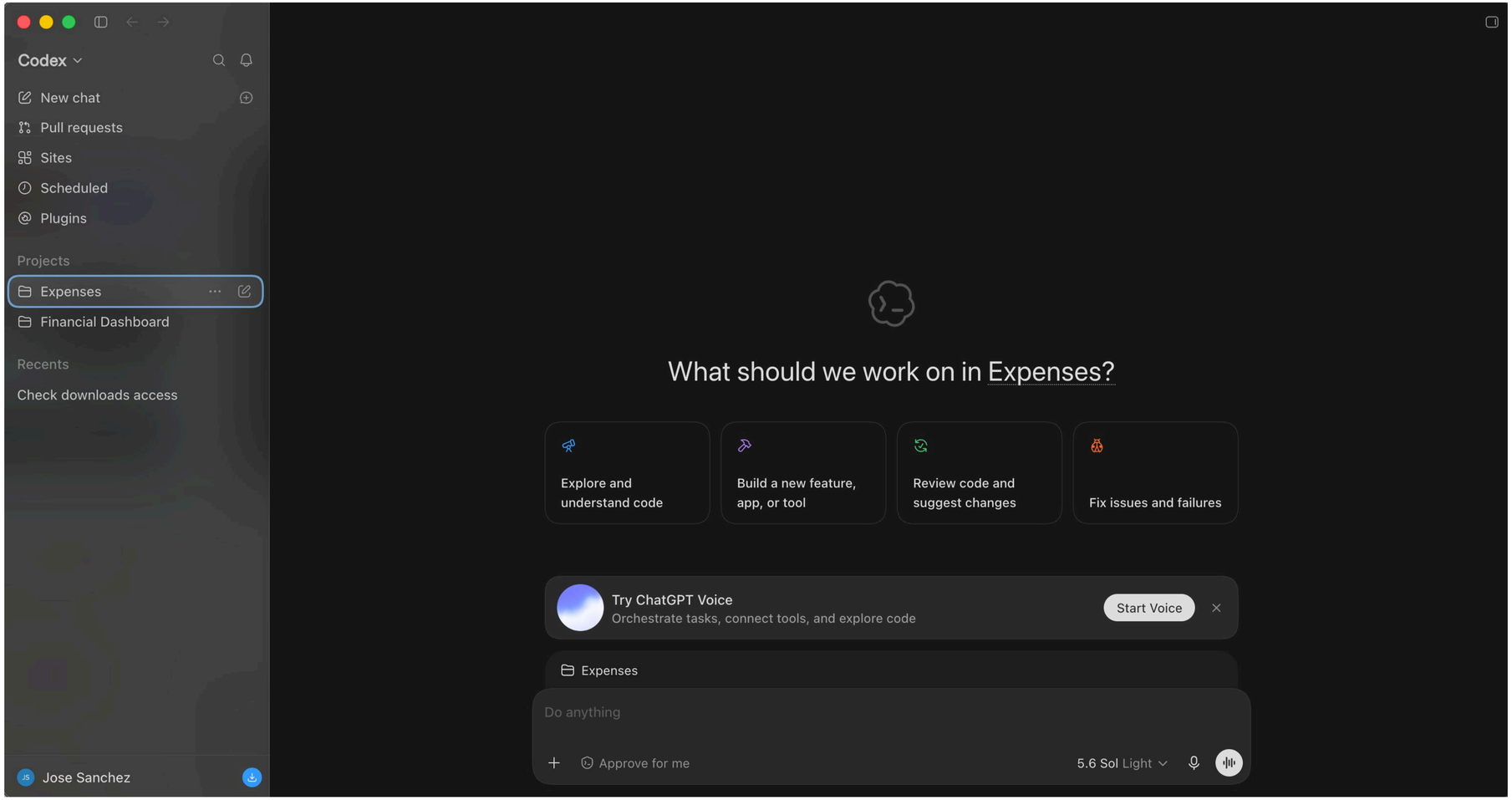

Tools
OpenAI Codex

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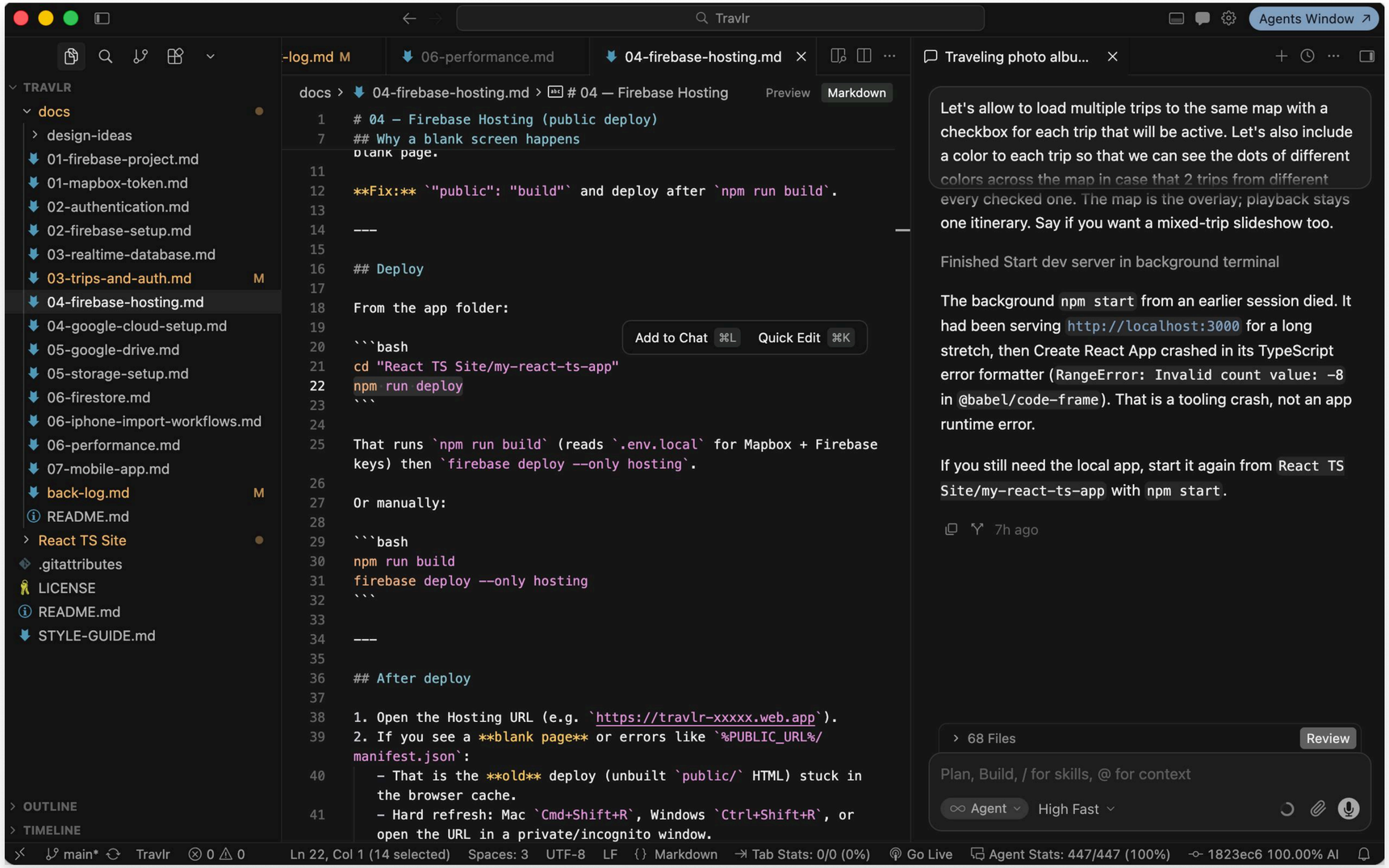
CURSOR



Tools

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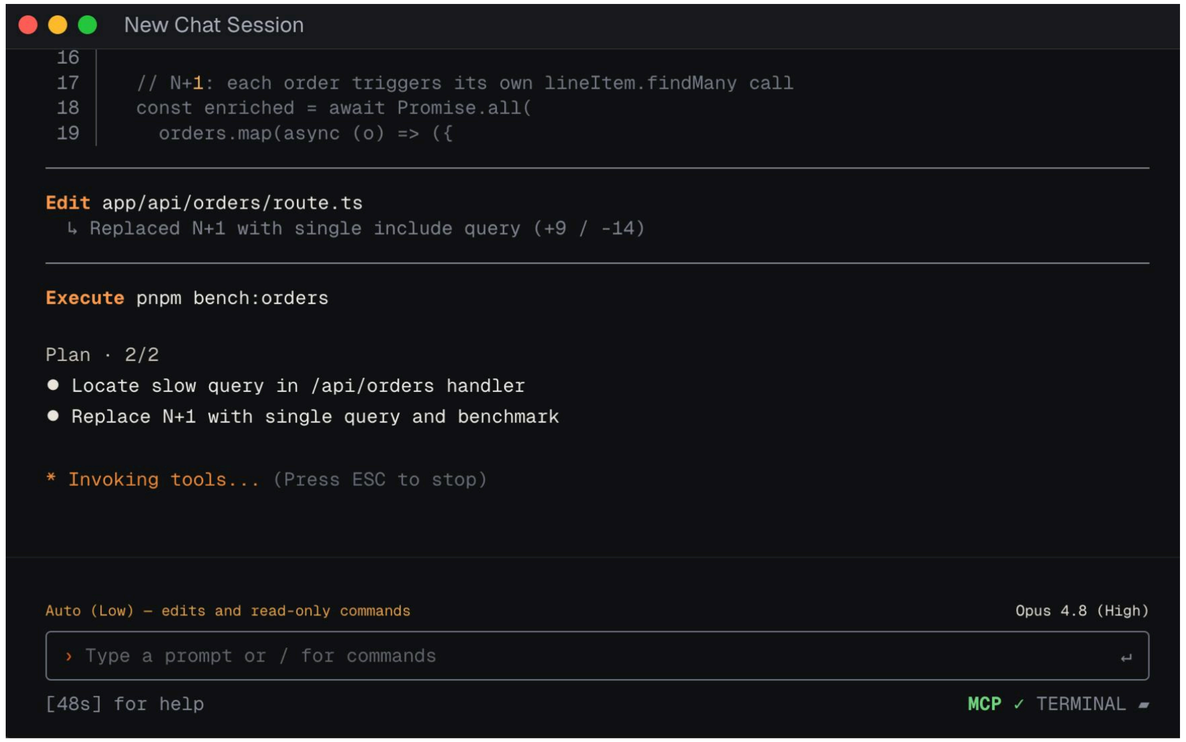

Tools

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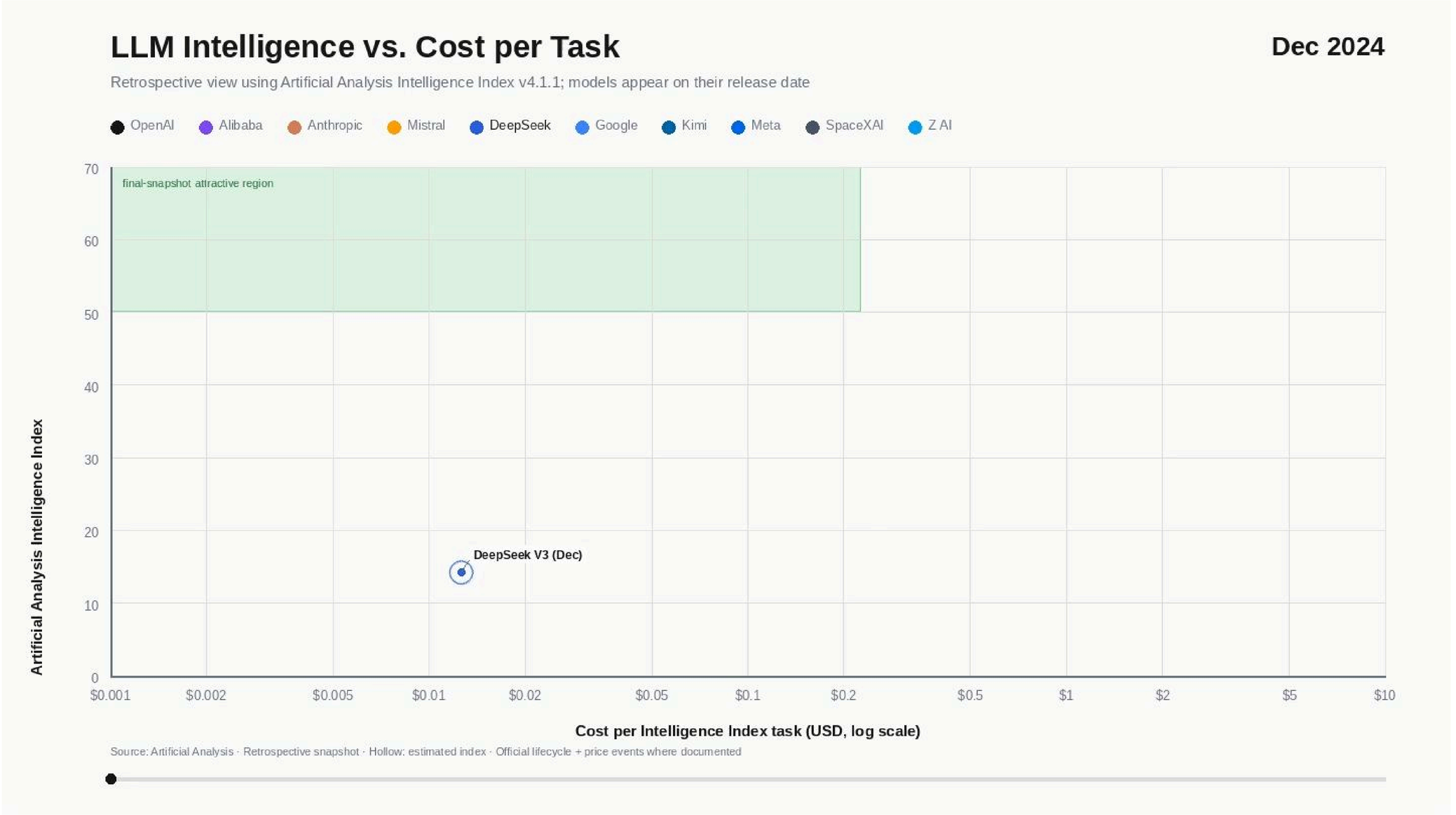
LLM Models



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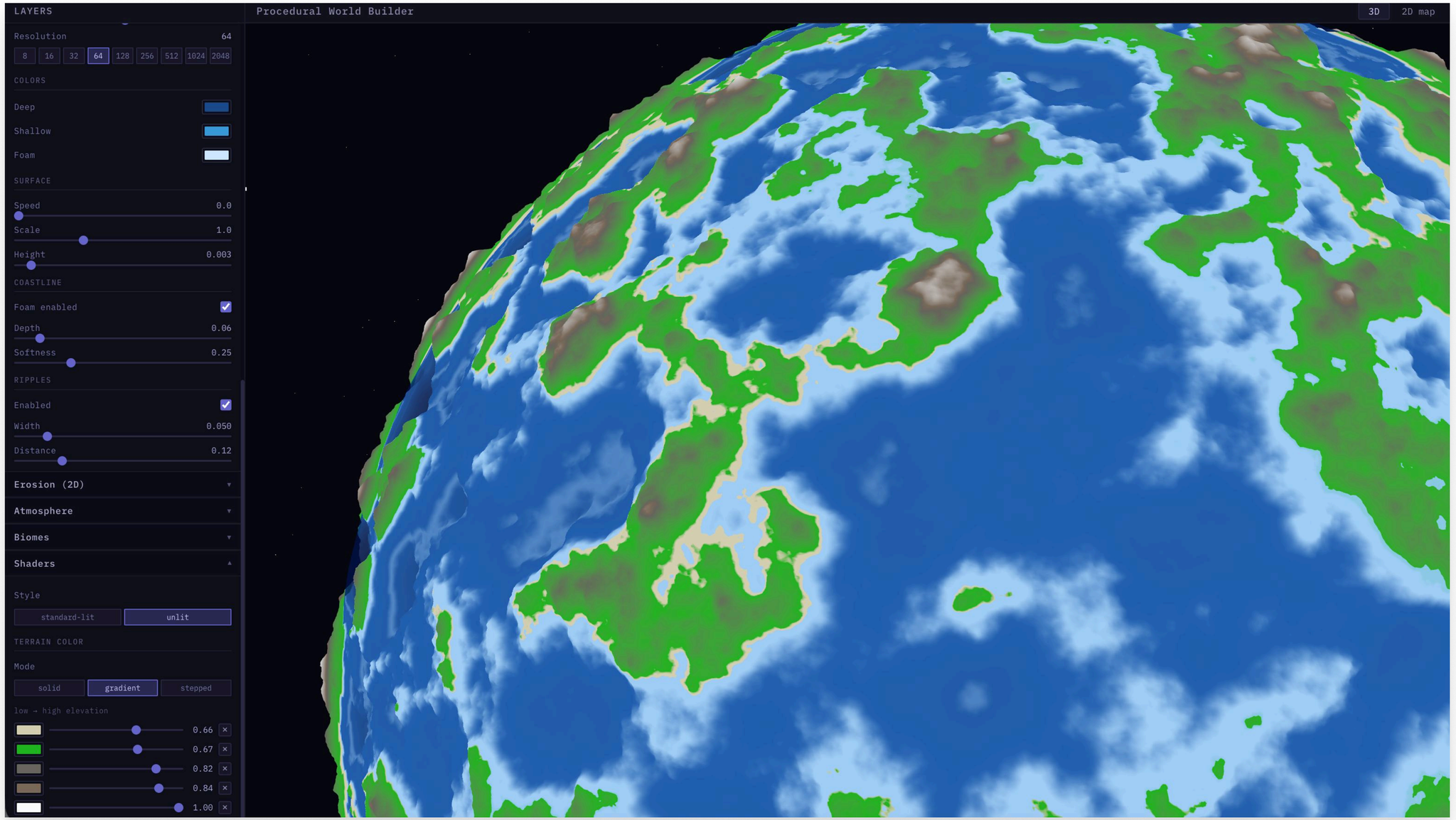
Experiments



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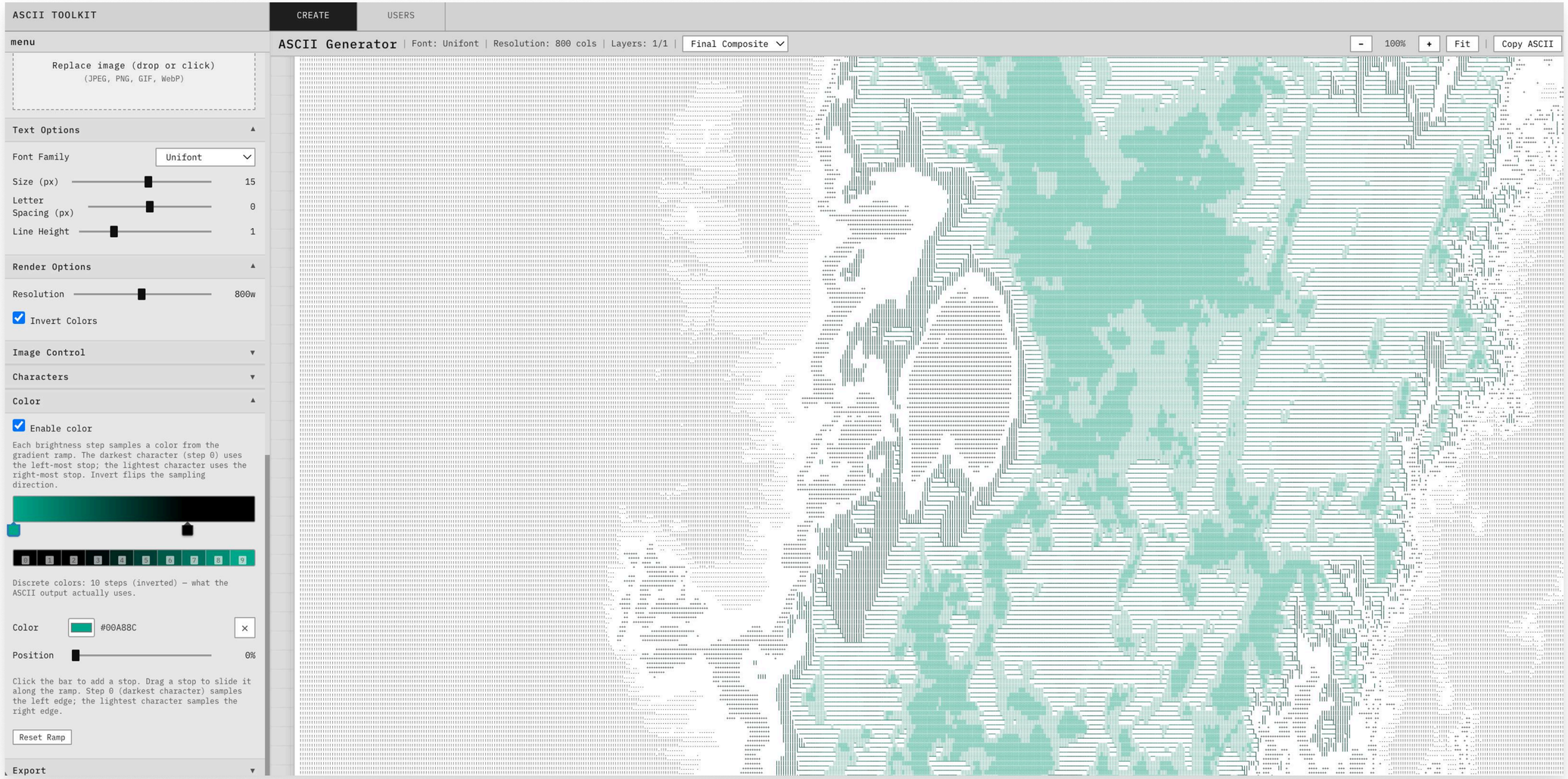
Experiments



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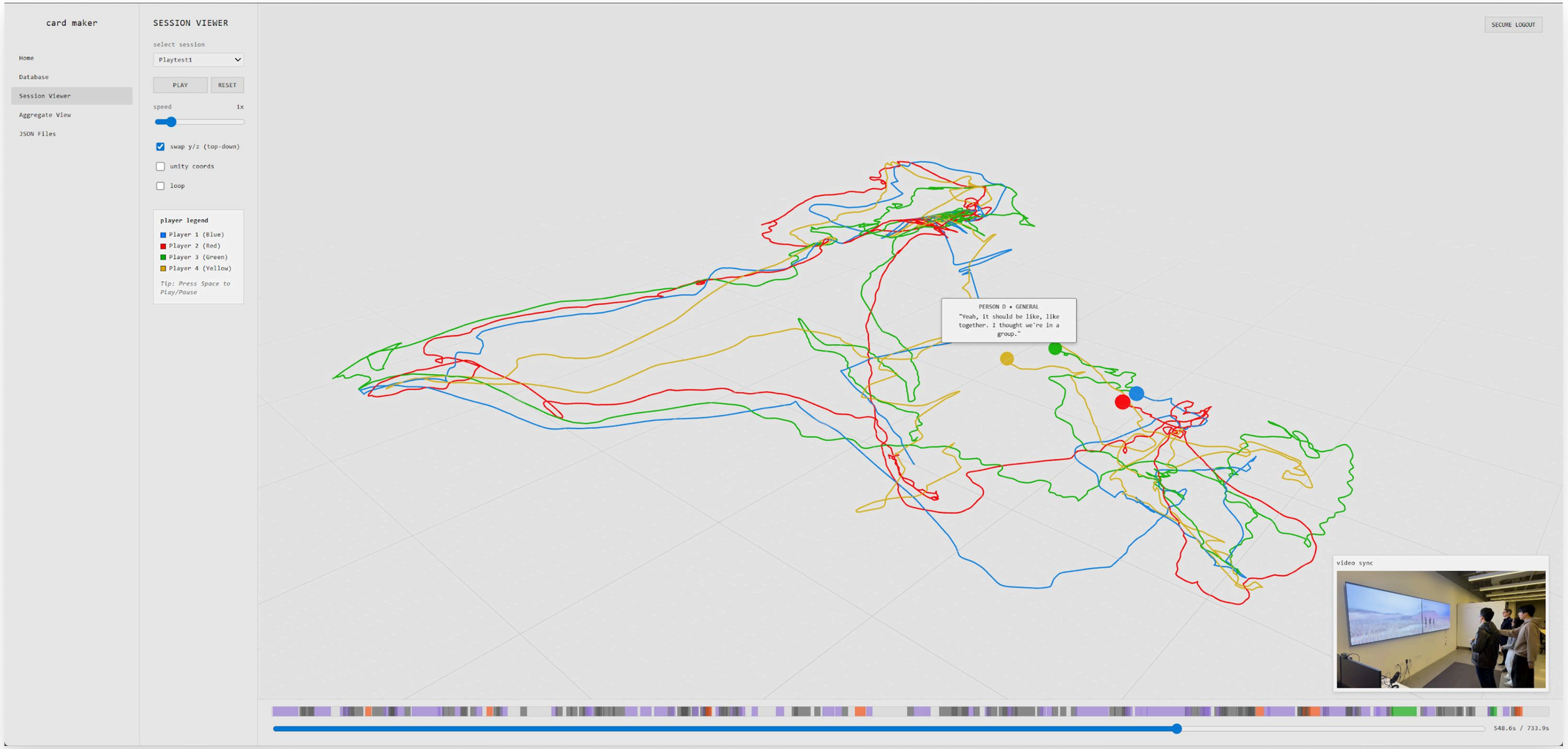
Experiments



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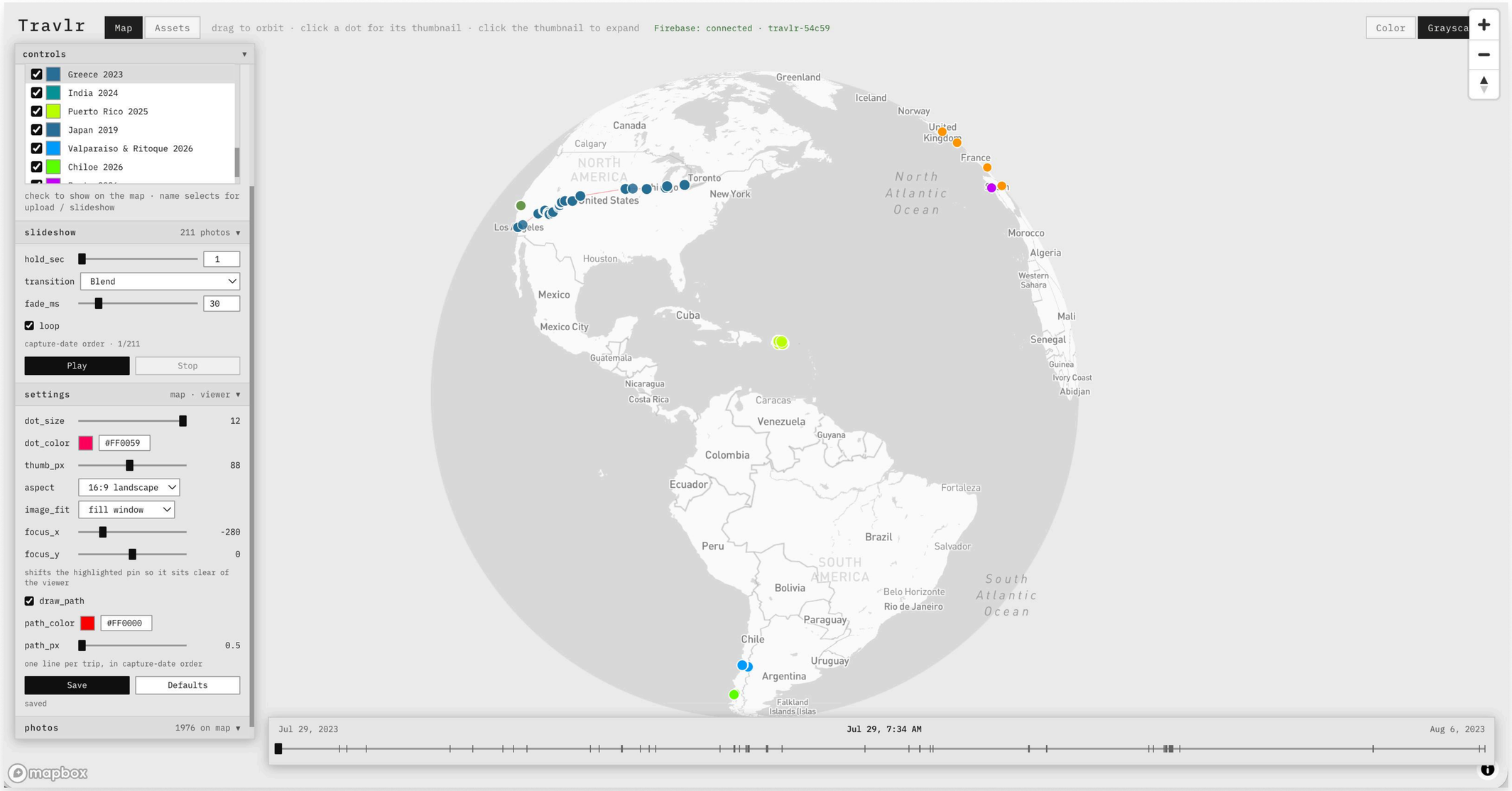

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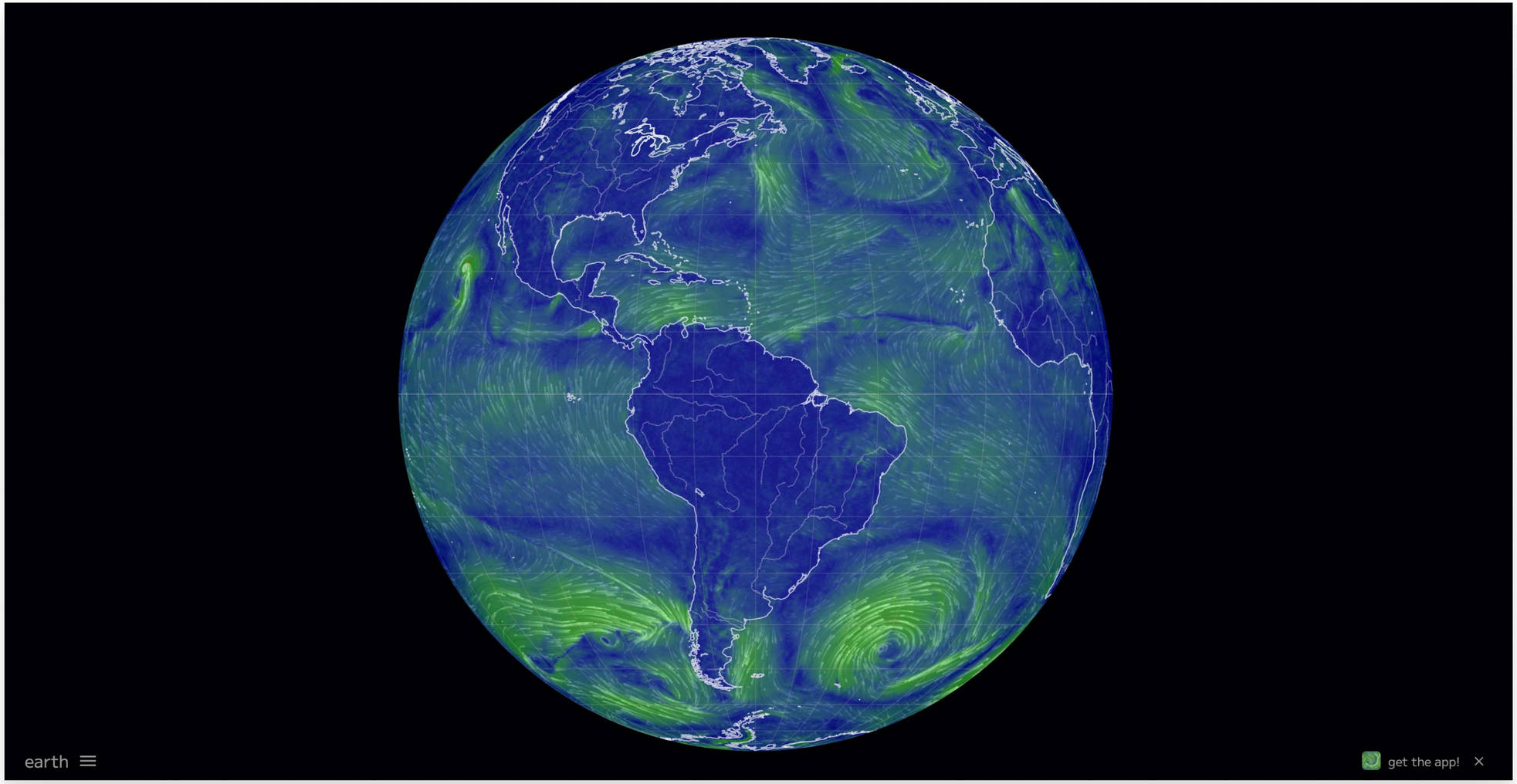

Experiments:
https://earth.nullschool.net/
Cameron Beccario

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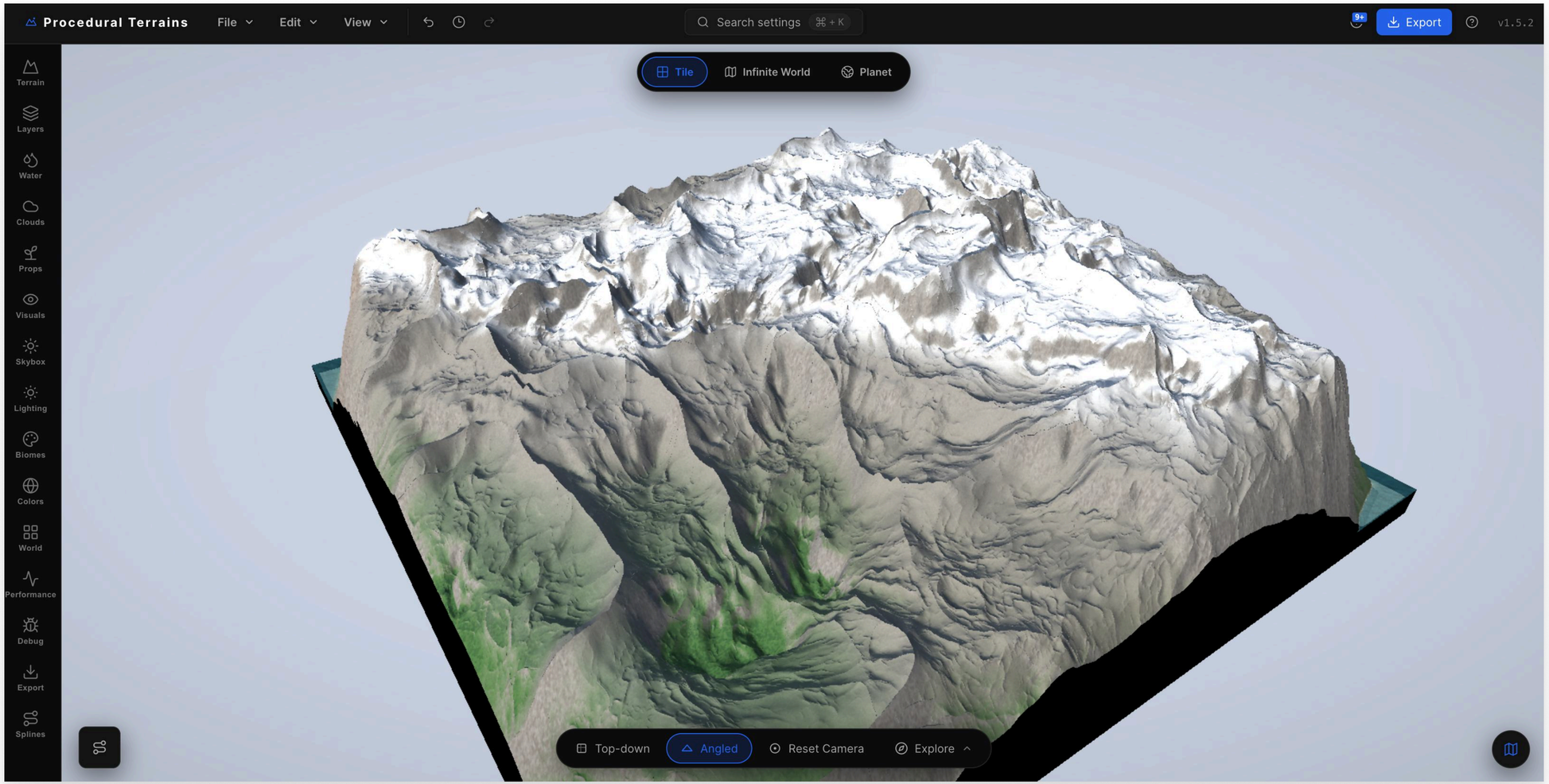

Experiments:
https://terrains.zyfod.dev/

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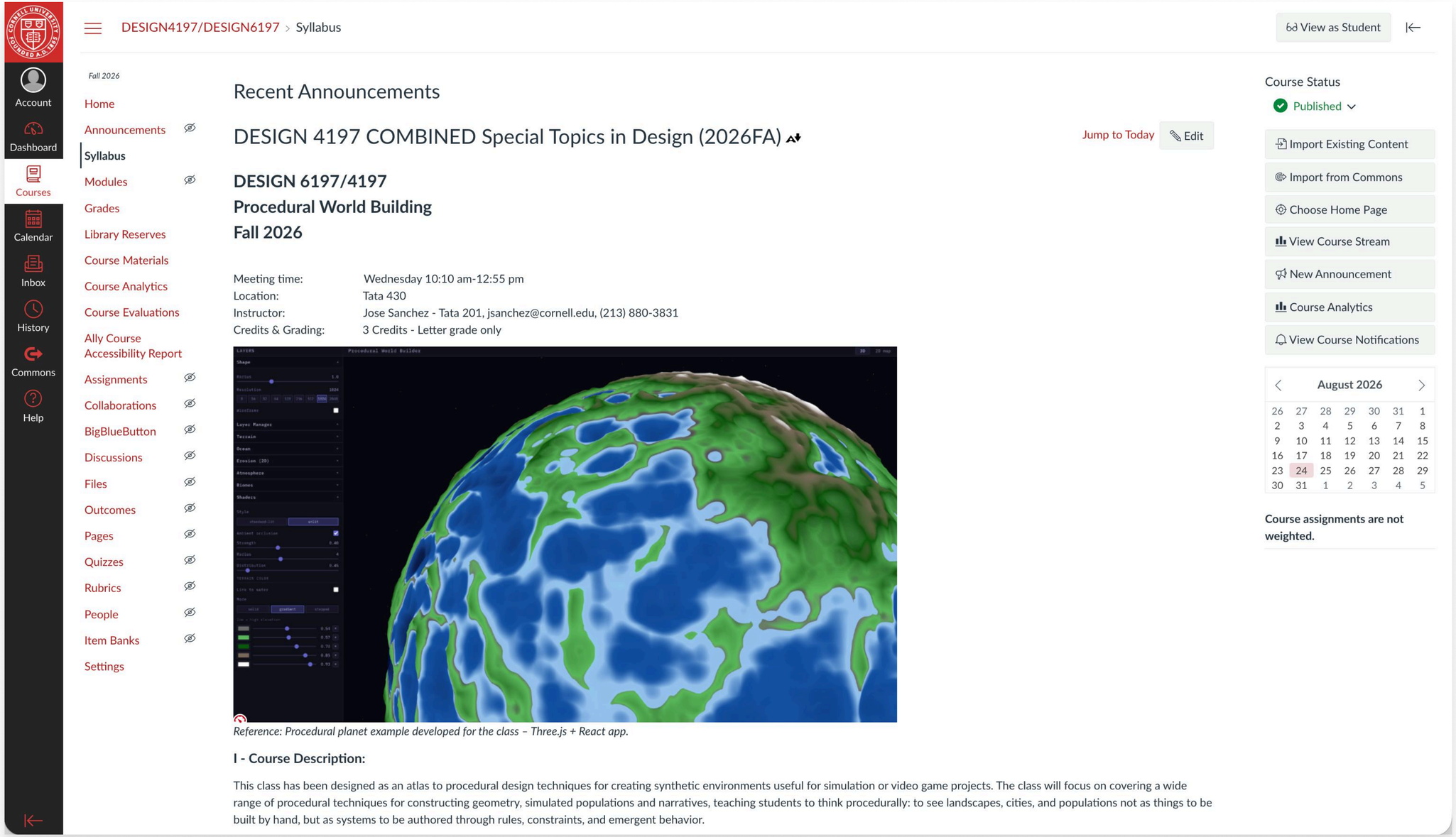
Syllabus



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Syllabus
Grade Rubric

		Excellent	Great	Good	Fair	Poor
Project ambition	Conceptual goals					
	Conceptual development					
	Ideas well executed					
Work Ethic	Makes effort to explain own goals/agenda					
	Reacts to constructive criticism					
	Supports classmates					
	Self-critical of project					
	Self motivated					
	Works what is expected					
	Completes work on time					
	Manages time in class					
	Asks instructor questions					
	Makes effort to meet with instructor during class					
Design	Design Position					
	Avoidance of Design Tropes					
	Use of Precedents					
	Originality					
	Transformation of Precedents					
	Aesthetic Intentionality					
Research	Computer modeling ability					
	Research ability					
	Documentation					
	Methodology					
Overall	Final ideas executed					
	Visual Communication					
	Overall understanding of project					
	Presentation					

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Syllabus
Sign-up

Progress Presentation Sign-Up

Type your full name in an empty slot. Max 6 students per session. Exactly 3 presentations each.

Session #	Date	Topic	Slot 1	Slot 2	Slot 3	Slot 4	Slot 5	Slot 6	# Signed Up
3	Sept 16	Procedural topography – terrain meshes and heightmaps							0
4	Sept 23	Erosion simulation – biomes and terrain texturing							0
5	Sept 30	Water, atmosphere and shading							0
6	Oct 7	Procedural vegetation – L-systems – ecosystems							0
7	Oct 14	Procedural cities I – road networks and parcels							0
9	Oct 28	Procedural cities II – buildings and urban fabric							0
10	Nov 4	Agents and behavior – flocking and pathfinding							0
11	Nov 11	Digital populations – memory and state							0
12	Nov 18	Emergent narrative – Final project proposal							0
13	Dec 2	World-building work session – desk critiques							0
	TOTAL	*Target: 60 presentations (20 students × 3)*							**0**

Skipped dates (do not sign up): Sept 2, Sept 9 (intro) · Oct 21 (Midterm) · Dec 9 (week before Final) · Dec 16 (Final) · Nov 25 (Thanksgiving)

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Assignment 1 -
Initiate your Github Repo:

- Create a landing page (Main Readme file)
- Create a docs folder
- In docs - create a planning folder
- In docs - create a tutorials folder
- In docs - create a analysis folder
- Create a backlog of features you would like to implement
- Commit and push your repo
- Submit a link to your repo